Pranav Tripathi

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EDUCATION

University of California, Berkeley | Mechanical Engineering | Regents' & Chancellor's Scholar May 2028
4.0 GPA | Courses: Mechanics, Solidworks, Python/MATLAB, Data Structures, Calculus, Lin Alg/Diff Equations

EXPERIENCE

Frostbyte Technologies | *Co-Founder* June 2024 – Present
- Leading a startup team to develop an autonomous robot to clear ice from sidewalks and improve pedestrian safety
- Awarded $12,000 by Samsung, and partnered with Gates Foundation to make device policy-compliant for public use
- Fabricated passive snow funnel and salt tank (400+ sq. ft. coverage), with FEA and TopOpt for 28% weight reduction
- Coordinated with software team to integrate control modules such as a gyro, GPS, and camera on a Raspberry Pi 5

D&Y Nexus Labs | *Mechanical Engineering Consultant* August 2026 – Present
- Consulting on the design of a methane flux detection module that eliminates the need for manual gas sampling
- Reinforced module for IP54 water resistance through custom gasket, implemented pressure-equalization for temperature and humidity relief, and assisted with an IoT server for remote sensing with <5 second latency
- Converted designs into tolerance drawings and creating bill of materials (BOM) to maintain budget under $200

San Francisco Engineers Without Borders | *Engineering Volunteer* August 2026 – Present
- Collaborating on structural/fluid systems to provide clean water to the 6200+ residents of Tranquillo Centro, Panama
- Designing solar-powered pump systems for off-grid operation, to provide additional 7,000 L/hr water supply
- Assisting with structural support (retaining walls, pipeline crossings), working towards system lifespan of 20 years

Computer Science Council | *President* June 2023 – June 2026
- Led 40+ member team to organize 14 free computer science competitions and workshops for young local students
- Managed 3 teams for event logistics, advertising, and activity development, and reached 1,800+ students in two years

RESEARCH

Embodied Dexterity Group & General Motors (GM) | *Undergraduate Design Lead* August 2025 – Present
- Leading the design of, QUAD a robotic hand with 11 DoF and visuotactile sensing, under PhD Ben Davis and GM
- Applying kinematic principles to design novel QDD linkages in the thumb and index fingers for high dexterity, and an asymmetric linkage between the middle and ring fingers for greater rigidity and non-uniform grasping
- Improving DFM and DFA through 2D machined internals and optimized tolerances with lower fabrication costs

Boston University Robotics Lab | *Mechanical Engineering Intern* June 2025 – August 2025
- Created autonomous navigation algorithms using solely visual input from a camera under Prof. Roberto Tron
- Developed simulations of the optical flow (movement) of points along path boundaries using Numpy and Matplotlib
- Programmed an OpenCV optical flow algorithm that calculates offset from a path using inverse kinematics, and ROS2 feedback nodes with PID control for course-correction, resulting in <18% deviation from path centerlines

Lake Washington Tech Math Dept. | *Assistant Researcher* December 2025 – May 2026
- Optimized popular search algorithms to enable pathfinding via minimal computations under Prof. Choudhury
- Simulated shortest-path algorithms Dijkstra's and A* in Python using data structures such as queues and linked lists
- Created a novel algorithm, "Convergent Bidirectional A*" which performs 93% faster than Dijkstra's and 52% more than A*, by running A* from both ends of a search and actively prioritizing paths that bring the two ends closer

SKILLS

Mechanical: 3D Modelling, Kinematics, FEA, Machining, GD&T, DFM/DFA, Microsoft Office (Excel Certified)
Electronics: Programming (Python, Java, C++, MATLAB), Github, Circuitry & Sensors (Fritzing)
Robotics: ROS2, Computer Vision (Optical Flow, Projections), Controls (PID, Odometry), Signals, SLAM